

Lecture - 5

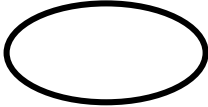
❖ Flowchart:

Flowchart is a diagrammatic/visual representation of an algorithm. A flowchart can be helpful for both writing programs and explaining the program to others. Flowcharts use symbols/shapes like arrows, rectangles, and diamonds to properly explain the sequence of steps involved in the algorithm or process.

➤ Symbols Used In Flowchart:

The below describes all the symbols that are used in making flowchart:

1. **Terminal/Terminator:** The oval symbol indicates Start, Stop and Halt in a program's logic flow. Terminal is the first and last symbols in the flowchart.

Symbol	Purpose	Description
	Terminal(Stop/Start)	Represents the start and the end of a flowchart.

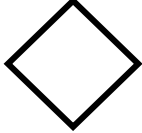
2. **Input/Output:** A parallelogram denotes any function of input/output type. Program instructions that take input from input devices and display output on output devices are indicated with parallelogram in a flowchart.

Symbol	Purpose	Description
	Input/output	Used for input and output operation.

3. **Action/Process:** A box represents arithmetic instructions, specific action or operation that occurs as a part of the process.

Symbol	Purpose	Description
	Processing	Used for arithmetic operations and data-manipulations.

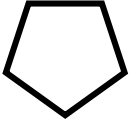
4. **Decision:** Diamond symbol represents a decision point. Decision based operations such as yes/no question or true/false are indicated by diamond in flowchart.

Symbol	Purpose	Description
	Processing	Used to represent the operation in which there are two alternatives, true and false.

5. **On-Page Connector:** Whenever a flowchart becomes complex, it is useful to use connectors to avoid confusion. An On-Page Connector shows the continuation of the flowchart on the same page and is represented by a small circle.

Symbol	Purpose	Description
	On-Page Connector	Used to join different flowchart on same page

6. **Off-Page Connector:** Whenever a flowchart spreads over more than one page, it is useful to use connectors to avoid confusion. An Off-Page Connector shows the continuation of the flowchart on a different page and is represented by a pentagon.

Symbol	Purpose	Description
	Off-Page Connector	Used to connect flowchart portion on different page.

7. **Flow Line:**

Arrows represent the direction of flow of control and relationship among different symbols of flowchart.

Symbol	Purpose	Description
	Flow Line	Used to indicate the flow of logic by connecting symbols.

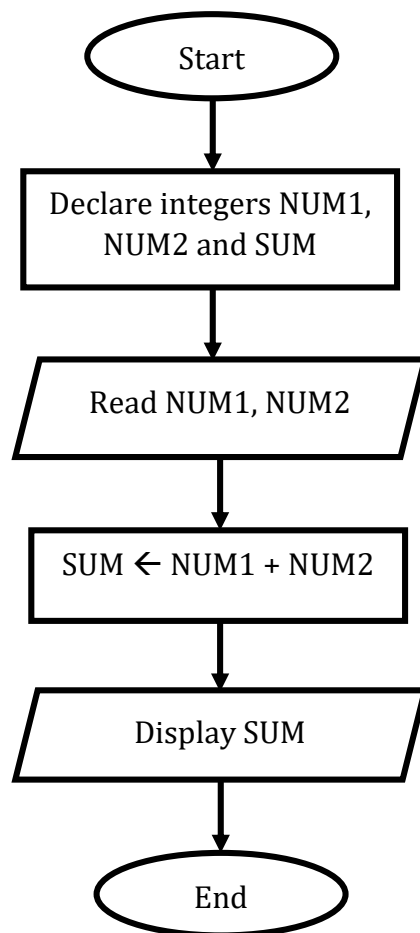
➤ **Standard Rules for Constructing a Flowchart:**

- ✓ **Clear Entry and Exit Points:** Every flowchart must begin with a single START terminal and end with an END or STOP terminal.
- ✓ **Directional Flow:** The standard flow of logic should move from top to bottom or left to right.

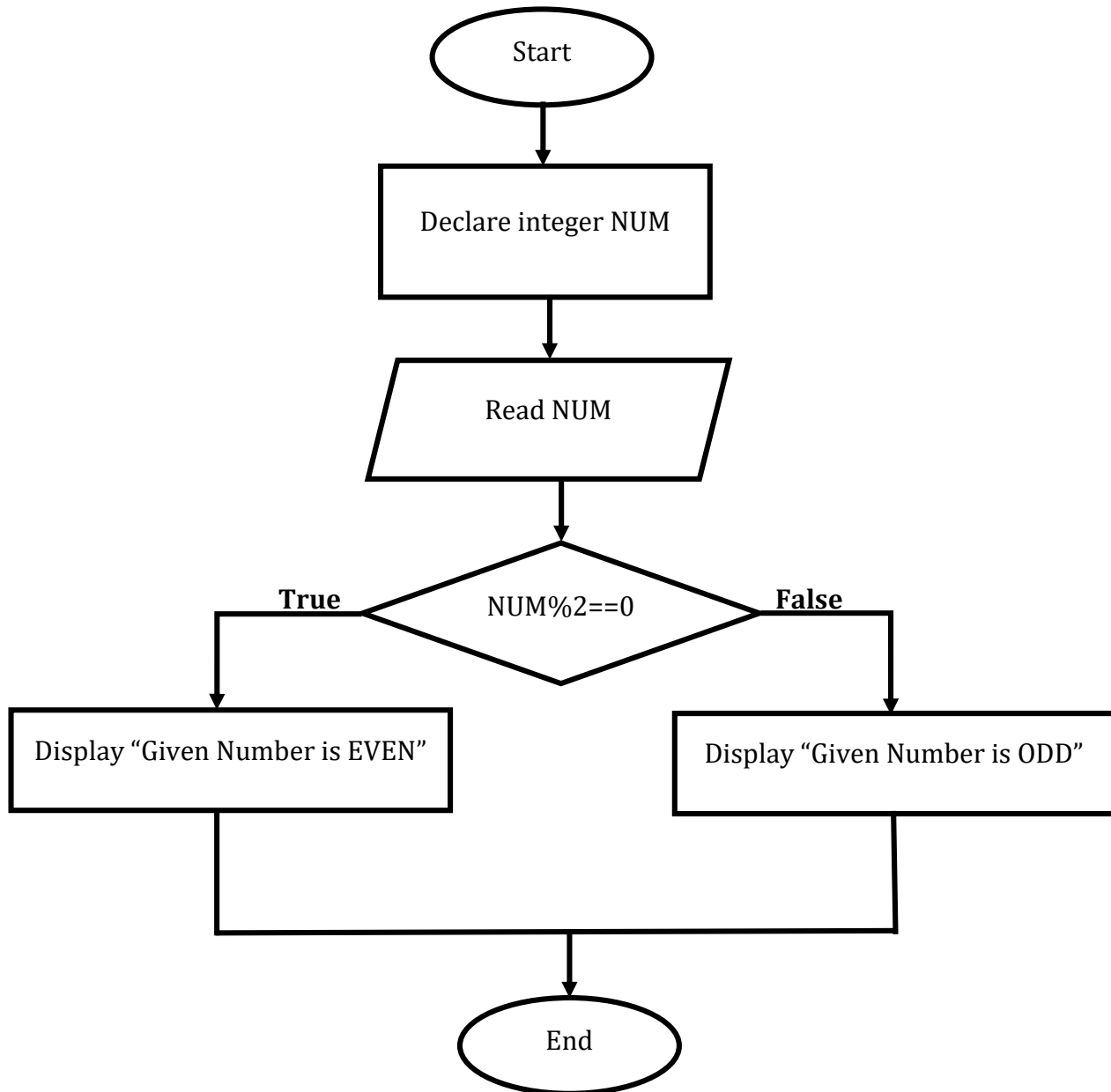
- ✓ **Consistent Arrow Lines:** All symbols must be connected using arrows (flow lines) to indicate the exact sequence of execution—never use plain lines. Arrowheads must point in the direction of the logic.
- ✓ **Avoid Crossing Lines:** Do not let flow lines intersect or cross each other. If it is unavoidable, use a small arc to show they cross without connecting.
- ✓ **Single Output for Processes:** A standard processing box should have only one exit flow line pointing to the next step.
- ✓ **Branching Clarity:** A decision symbol must have exactly one entry point and two or more distinct exit lines representing outcomes (e.g., True/False, Yes/No, or multiple cases for a switch statement).
- ✓ **Use Connectors for Multi-Page Diagrams:** If a flowchart becomes too complex or spans multiple pages, use circular on-page connectors or off-page connectors to cleanly bridge the gap without creating a mess of lines.

➤ **Example Flowcharts:**

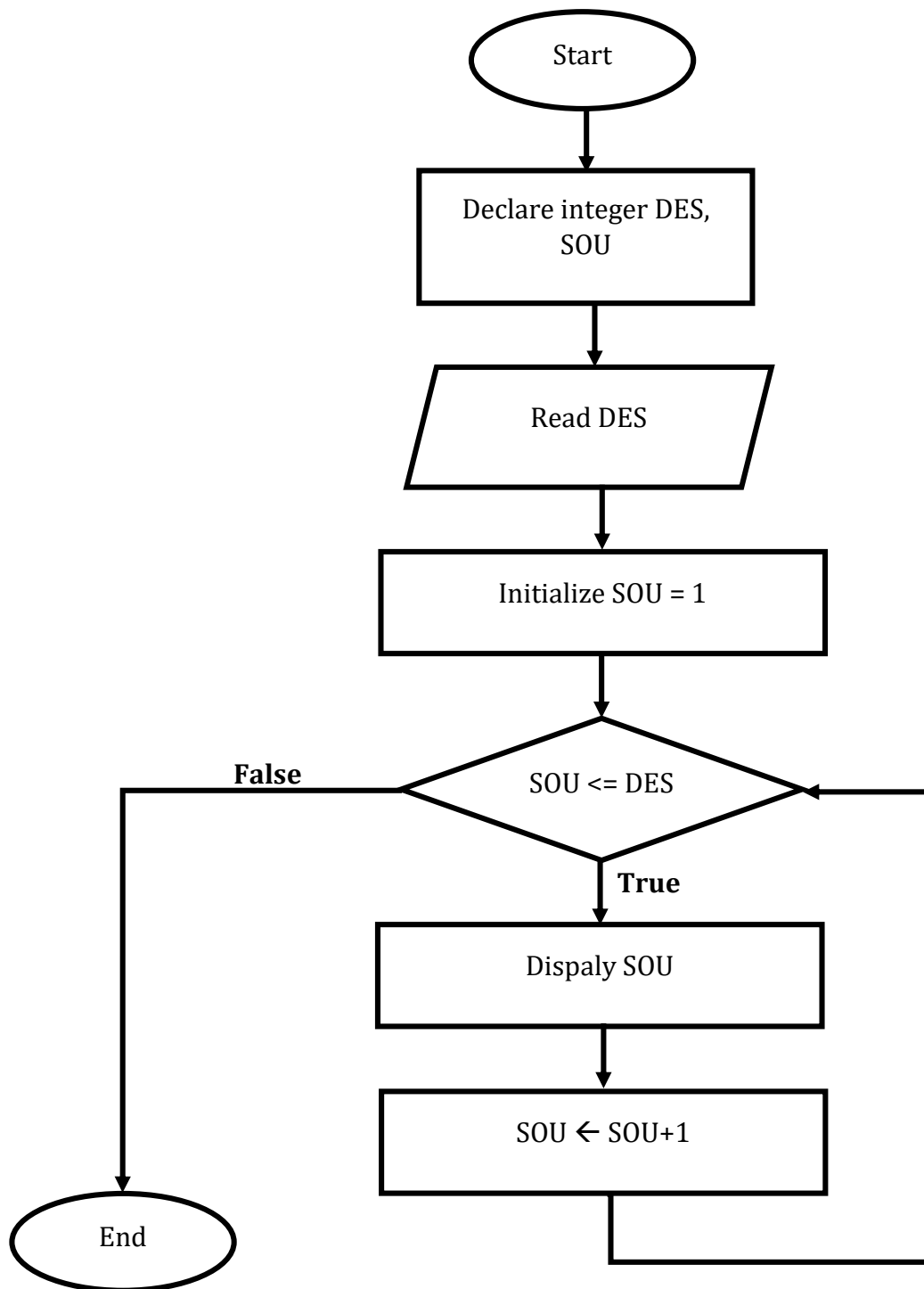
Exp. 1 : Design flowchart for addition of two numbers (Sequential)



Exp. 2 : Design flowchart to check EVEN or ODD (Selection/Decision)



Exp. 3 : Design flowchart to print N natural numbers (Repetition/Iteration/Looping)



Assignment:

1. Draw a flowchart to calculate the simple interest.
2. Draw a flowchart to find the average of three numbers.
3. Draw a flowchart to calculate the total and percentage of marks in five subjects.
4. Draw a flowchart to check whether a number is positive, negative, or zero.
5. Draw a flowchart to find the largest of three numbers.
6. Draw a flowchart to check whether a given year is a leap year.
7. Draw a flowchart to check whether a person is eligible to vote.
8. Draw a flowchart to find the roots of a quadratic equation.
9. Draw a flowchart to find the sum of the first N natural numbers.
10. Draw a flowchart to find the factorial of a number.
11. Draw a flowchart to reverse a number.
12. Draw a flowchart to check whether a number is a palindrome.
13. Draw a flowchart to check whether a number is an Armstrong number.
14. Draw a flowchart to find the sum of even and odd numbers separately from 1 to N.
15. Draw a flowchart to print all prime numbers between 1 and N.

Sample Important Questions:

1. State the purpose of the decision symbol.
2. List the basic flowchart symbols.
3. What does the terminal symbol represent?
4. Explain the purpose of arrows in a flowchart.
5. Explain the purpose and importance of flowcharts in problem solving.
6. Explain the commonly used flowchart symbols with their functions.
7. Differentiate between sequence, selection, and iteration constructs in flowcharts.
8. Explain the difference between an algorithm and a flowchart with a suitable example.